


```
mamba install seaborn scikit-learn
mamba env export --from-history > environment.yml
conda-lock lock -f environment.yml
```

Finding packages

- (1.) Search for a package by name from the command line:

```
mamba search numpy
mamba search r-ggplot2
mamba search samtools
```

- (2.) If you prefer browsing, use the channel web interfaces.

>> Sharing and managing environment files

- (1.) When working with Git to track changes  SOP 0101, commit both `environment.yml` and the lockfile to the project repository alongside the source code. When you change one, regenerate and commit the other in the same commit.

Critical: Updating `environment.yml` without regenerating the lockfile leaves the two artifacts inconsistent: the next collaborator who installs from the lockfile will not see your changes; the next who installs from `environment.yml` will get a different version set than you tested.

- (2.) Pin top-level packages with major and minor versions in `environment.yml`; leave patches floating.
- (3.) To recreate an environment exactly from a lockfile on a new machine, install from the lockfile rather than the environment file.

```
conda-lock install --name my-project conda-lock.yml
```

- (4.) When working with an external repository, apply the same discipline. If forking to modify or extend: adopt the upstream `environment.yml` if one exists (create one if not), then immediately regenerate a lockfile in your fork. If cloning without modification: create the environment from the upstream file as-is.

```
mamba env create -f environment.yml
mamba activate my-project
```

Updating and pruning

- (1.) Update mamba periodically. Update only in base, not in project environments.

```
mamba update -n base mamba
```

- (2.) List existing environments and remove ones for projects no longer worked on.

```
mamba env list
mamba env remove -n old-project
```

- (3.) Reclaim disk space from the package cache.

```
mamba clean --all
```

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